

NIELIT Virtual Academy

COURSE PROSPECTUS

Name of the Course: O level - Programming and Problem Solving through Python Language

Course Code: ITB43

Mode of Conduction: Online[Self-Paced]

Duration: 120 Hours

Objective of the Course

The objectives of this module are to make the learners understand the programming language concepts like Data Types, Loops, Functions; Python Lists, Strings, Tuples, Dictionaries, Elementary Data Handling using Pandas, NumPy etc.

After completion of this course, the learner is expected to analyze the real life problem and write a program in Python to solve the problem.

The main emphasis of the module will be on writing algorithm to solve problems and implement in Python. After completion of the module, the learner will be able to

- ✓ Draw flow charts for solving different problems.
- ✓ Develop efficient algorithms for solving a problem.
- ✓ Use the various constructs of Python viz. conditional, iteration.
- ✓ Write programs making judicious use of Lists, Strings, Tuples, and Dictionaries wherever required.
- ✓ Manage data using NumPy.
- ✓ Handle files and create Modules in Python.

Outcome of the Course:

Python is the fastest growing programming language globally, mainly due to its simplicity and large community. Python is easy to use, powerful and versatile programming language, making it a great choice for programmer/developer. This course introduces programming and the Python language. Students are introduced to core programming concepts and the course content theory and practical both so that students understand concept and code to implement the program in a organized way. Also various tools available for writing and running Python programs with coding examples and assignments..

After completing the module, the learner will be able to:

- ✓ Understand the python Language and describe its features.
- ✓ Understand the concept of Python Program and make program by using Control Structure.
- ✓ Able to understand the User Defined Function and create the UDF.
- ✓ Able to understand Sequence Data Type (List, Tuple, Set and Dictionary).
- ✓ Understand the concept of Module/ Package in python.
- ✓ Understand the File Handling in Python.
- ✓ Understand working with Multidimensional Array Data By Using Numpy.
- ✓ Able to learn Machine Learning(ML) or AI Using Python

Course Fee: Rs: 590/-

Eligibility:

12th

Or ITI Certificate (Two Years) after class 10

Or ITI Certificate (One Years) after class 10 with one year of experience post qualification. Or Successful completion of the second year of a Government recognized polytechnic engineering diploma course after class 10, Training of 'O' Level course concurrently during the third year of the said 3 years Polytechnic engineering diploma course.

Methodology:

- ✓ Teaching Mode: Self-Pace
- ✓ Access from anywhere anytime
- ✓ Content Access through e-learning portal
- ✓ Doubt Clearing Session
- ✓ Practical Oriented

Registration Link: <http://nva.nielit.gov.in>

Contact Details:

- Course coordinator Name: Sh. Raja Ram
- Email: rajaram@nielit.gov.in
- Mobile number: 7706009313

Contents:

Module: M3-R5.1 Python	
Introduction to Programming	
UNIT 1	The basic Model of computation, algorithms, flowcharts, Programming Languages, compilation, testing & debugging and documentation.
Algorithms and Flowcharts to Solve Problems	
UNIT 2	Flow Chart Symbols, Basic algorithms/flowcharts for sequential processing, decision based processing and iterative processing. Some examples like: Exchanging values of two variables, summation of a set of numbers, Decimal Base to Binary Base conversion, Reversing digits of an integer, GCD (Greatest Common Divisor) of two numbers, Test whether a number is prime, factorial computation, Fibonacci sequence, Evaluate 'sin x' as sum of a series, Reverse order of elements of an array, Find largest number in an array, Print elements of upper triangular matrix, etc.
Introduction to Python	
UNIT 3	Python Introduction, Technical Strength of Python, Introduction to Python Interpreter and program execution, Using Comments, Literals, Constants, Python's Built-in Data types, Numbers (Integers, Floats, Complex Numbers, Real, Sets), Strings (Slicing, Indexing, Concatenation, other operations on Strings), Accepting input from Console, printing statements, Simple 'Python' programs.
Operators, Expressions and Python Statements	
UNIT 4	Assignment statement, expressions, Arithmetic, Relational, Logical, Bitwise

	operators and their precedence, Conditional statements: if, if-else, if-elif-else; simple programs, Notion of iterative computation and control flow –range function, While Statement, For loop, break statement, Continue Statement, Pass statement, else, assert.
Sequence Data Types	
UNIT 5	Lists, tuples and dictionary, (Slicing, Indexing, Concatenation, other operations on Sequence datatype), concept of mutability, Examples to include finding the maximum, minimum, mean; linear search on list/tuple of numbers, and counting the frequency of elements in a list using a dictionary.
Functions	
UNIT 6	Top-down approach of problem solving, Modular programming and functions, Function parameters, Local variables, the Return statement, DocStrings, global statement, Default argument values, keyword arguments, VarArgs parameters. Library function-input(), eval(), print(), String Functions: count(), find(), rfind(), capitalize(), title(), lower(), upper(), swapcase(), islower(), isupper(), istitle(), replace(), strip(), lstrip(),rstrip(), split(), partition(), join(), isspace(), isalpha(), isdigit(), isalnum(), startswith(), endswith(), encode(), decode(), String: Slicing, Membership, Pattern Matching, Numeric Functions: eval(), max(), min(), pow(), round(), int(), random(), ceil(), floor(), sqrt(), Date & Time Functions, Recursion.
File Processing	
UNIT 7	Concept of Files, File opening in various modes and closing of a file, Reading from a file, Writing onto a file, File functions-open(), close(), read(), readline(), readlines(), write(), writelines(), tell(), seek(), Command Line arguments.
Scope and Modules	
UNIT 8	Scope of objects and Names, LEGB Rule Module Basics, Module Files as Namespaces, Import Model, Reloading Modules.
NumPy Basics	
UNIT 9	Introduction to NumPy, ndarray, datatypes, array attributes, array creation routines, Array From Existing Data, Array From Numerical Ranges, Indexing & Slicing.

Examination & Certification

Category	Theory	Total
M3-R5.1: Programming and Problem Solving through Python Language	01	100
Marks	100	100

After successful completion of the course, candidate will get an online certificate with the following Grading Scheme:

Marks Range	Grade	Certificate Type
85% and above	S	Graded
75-84%	A	Graded
65-74%	B	Graded
55-64%	C	Graded
50-54%	D	Graded
<50%	F	Participation
Attended the Course but not appeared in Examination	N	Participation